AAV Titer Quantification

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SUSTech-BIO 2026 iGEM Team

Purpose

Determine the physical genome titer of recombinant AAV using the Beyotime Universal AAV Titration Kit by qPCR (C2921S) on a Bio-Rad CFX96. The assay targets AAV2-derived ITRs and reports vector genomes per milliliter (vg/mL); it does not measure infectious or functional titer. [1]

Materials & Equipment

Item	Supplier / identifier	Use / specification
Universal AAV Titration Kit by qPCR	Beyotime C2921S	50 reactions at 20 µL; use the supplied extraction reagents, qPCR mix, primer mix, Positive Control and water
Recombinant AAV sample	Preparation from AA02	AAV2-derived ITRs; retain the original sample and all dilution records
Real-time PCR system	Bio-Rad CFX96	SYBR Green detection; no ROX required [1]
Thermal cycler or heating blocks	Laboratory equipment	37°C, 55°C and 95°C for sample pretreatment
qPCR plates or tube strips; optical seals	CFX96-compatible consumables	Nuclease-free; use the plate type configured in the instrument
Pipettes; filter tips; tubes	Laboratory supplies	Nuclease-free liquid handling and serial dilutions
Ice; vortex mixer; centrifuge	Laboratory equipment	Reagent cooling, mixing and brief plate/tube centrifugation

Before You Start

Confirm that the vector contains AAV2-derived ITRs; capsid serotype alone does not determine compatibility. Start with an AAV preparation from AA02 and identify it as purified or crude, because the DNase I incubation differs. [1]

Thaw and gently mix kit reagents before use; keep qPCR reagents on ice and protect the fluorescent mix from light. Store the kit at -20°C and minimize repeated freeze-thaw cycles. Keep the concentrated Positive Control separate from sample preparation. [1]

Plan a standard curve and a water no-template control (NTC) on every plate, with technical replicates for standards and samples. Triplicates are a practical starting design; prioritize additional replicates for low-copy standards when extending the curve. Include an extraction blank when checking contamination introduced during pretreatment.

Record any dilution of intact virus before extraction as Dpre, and the total dilution of the final lysate before qPCR as Dpost. The kit recommends at least a fivefold lysate dilution (Dpost ≥ 5). Budget standards, controls, sample replicates and pipetting excess within the kit capacity. [1]

Procedure

A. Remove free DNA and release AAV genomes

1. For each sample, assemble the following 20 µL DNase I reaction using 2 µL of the AAV preparation. Record any prior virus dilution. Process an extraction blank in parallel by replacing the AAV input with water, if included. [1]

Component	Volume per sample
AAV preparation	2 µL
Reaction Buffer (10X)	2 µL
Ultrapure Water	14 µL
DNase I (1 U/µL)	2 µL
Total	20 µL

2. Mix and incubate at 37°C for 15 min for purified AAV, or 60 min for crude AAV obtained by freeze-thaw preparation. Heat at 95°C for 10 min to inactivate DNase I. Use a heated lid to limit evaporation. [1]

3. Freshly combine 45.5 µL DNA Extraction Solution and 2 µL Enzyme Mix per sample. Mix well, then add 5 µL of the DNase-treated, heat-inactivated sample to this 47.5 µL digestion mixture. [1]

4. Incubate at 55°C for 15 min, then at 95°C for 5 min. Add 47.5 µL Stop Solution and vortex to mix. The resulting genome lysate volume is 100 µL. Use immediately or store at -20°C. [1]

5. Dilute the lysate at least fivefold with supplied Ultrapure Water before qPCR. For example, combine 5 µL lysate with 20 µL water for Dpost = 5; scale to the number of wells needed. If inhibition is suspected, test a further dilution and compare dilution-corrected results. [1]

B. Prepare the Positive Control standard curve

6. Check the Positive Control concentration on the supplied label. For the 2 × 10^9 copies/µL stock specified in the IFU, prepare ST0-ST5 as below, mixing each tube thoroughly before transferring to the next and changing tips at every transfer. Use 5 µL of each dilution per qPCR well. [1]

Standard	Preparation	copies/µL	copies/reaction
ST0	5 µL stock + 45 µL water	2 × 10^8	10^9
ST1	5 µL ST0 + 45 µL water	2 × 10^7	10^8
ST2	5 µL ST1 + 45 µL water	2 × 10^6	10^7
ST3	5 µL ST2 + 45 µL water	2 × 10^5	10^6
ST4	5 µL ST3 + 45 µL water	2 × 10^4	10^5
ST5	5 µL ST4 + 45 µL water	2 × 10^3	10^4

7. If low titers are expected, add Positive Control dilutions at the low-concentration end so that the intended standard curve brackets the samples. For example, continue the same 1:10 dilution from ST5 to obtain 10^3 copies/reaction and, if needed, another point at 10^2 copies/reaction. These are optional extension points, not a manufacturer-validated lower quantification range.

Accept an extended low-copy point only when replicate amplification is reproducible, melt profiles are consistent with the standards, and the signal is distinguishable from the NTC. Review the low-end fit as well as the overall regression; a high overall R^2 alone does not validate the added points. Keep failed or inconsistent extension points out of the accepted quantitative range and document why.

C. Assemble qPCR reactions

8. Prepare a common master mix without template, allowing for pipetting loss. Dispense 15 µL per well and add 5 µL of diluted sample, standard or water for the NTC. Do not put the plasmid Positive Control through the DNase I pretreatment. [1]

Component	Volume per 20 µL reaction
BeyoFast SYBR Green qPCR Mix (2X, with UDG)	10 µL
AAV Primer Mix (10 µM)	1 µL
Ultrapure Water	4 µL
Template: diluted lysate, standard or NTC water	5 µL
Low ROX / High ROX	None on CFX96
Total	20 µL

9. Mix gently, seal the plate or tubes, and centrifuge briefly to collect liquid and remove bubbles. Verify the physical plate against the well map, including standard quantities, dilutions and replicate labels. [1]

D. Run the CFX96

10. Select SYBR Green detection, set a 20 µL reaction volume and enter the kit cycling conditions below. Assign standards by copies per reaction, rather than copies per microliter, to match the calculation in section E. [1]

Stage	Temperature	Time / repeats	Read
UDG treatment	50°C	5 min; once	No
Initial denaturation	95°C	2 min; once	No
Denaturation	95°C	15 s	No
Annealing / extension	60°C	30 s	Yes
Repeat the last two stages	-	40 total cycles	At each 60°C step
Melt curve	CFX melt program	Inspect range and increments	At every increment

In a GOTO editor, return to the denaturation step 39 additional times for 40 cycles in total. The IFU lists an optional melt analysis using a QuantStudio example. On CFX96, append a native melt-curve acquisition step and save its temperature range, increment and hold time; three temperature holds alone do not replace a scanned melt curve. [1]

11. Check that fluorescence collection is enabled at the end of each 60°C step and during melting, then start the run. Save the thermal protocol, well map and original run file.

E. Check the run and calculate genome titer

12. Inspect amplification curves, melt curves and replicate agreement before fitting the standard curve. Review the NTC and any extraction blank first. The IFU gives no typical amplification or Cq > 34 as its NTC reference, but this alone is insufficient to validate low-copy extension points near background. [1]

13. Fit Cq = m × log10(Q) + b using accepted standards, where Q is copies per reaction. Record the slope, intercept, R^2 and efficiency, calculated as E = (10^(-1/m) - 1) × 100%. Use consistent baseline and threshold settings within the assay and document any excluded wells.

14. For a sample within the accepted standard range, calculate Q = 10^((Cq - b)/m), then convert to the original-virus genome titer using the expression below. Apply each dilution factor once.

Titer (vg/mL) = Q × 200 × Dpost × Dpre × 1000 / 5 = Q × 40,000 × Dpost × Dpre. [1]

The fixed factor 200 comes from the extraction volumes: 2 µL virus in 20 µL, followed by transfer of 5 µL into a final 100 µL lysate. Dpost is the subsequent lysate dilution; Dpre is any intact-virus dilution before extraction (1 if none). The denominator 5 is the qPCR template volume in µL. If software reports copies/µL instead, reconcile the quantity units before using this formula.

15. Compare dilution-corrected titers across valid sample dilutions and summarize accepted technical replicates. If a sample falls below the lowest reliable standard, report it as below the

established quantification range and repeat with validated lower standards. Any provisional extrapolated value must be labeled as an extrapolation, not a confirmed titer.

16. Retain the raw run file, Cq export, standard quantities, amplification and melt curves, fit statistics, replicate exclusions, dilution calculations and the final result in vg/mL. Do not interpret technical replicates as independent virus preparations.

p.s.

For low-titer AAV, plan extra low-concentration Positive Control gradients before running the plate. The aim is to bracket the sample with reliable standard points; simply adding more high-concentration controls will not resolve low-end extrapolation.

Extra standard points do not automatically increase assay sensitivity. If their signals overlap the NTC or vary substantially between replicates, the lower range has not been established.

Keep the kit's recommended lysate dilution in mind when signal is weak: using less-diluted lysate may also increase inhibition. Check dilution behavior before changing template input or pretreatment volumes. [1]

DNase I removes free DNA before genome extraction. Skipping it can overestimate the titer through residual packaging plasmid; the kit's UDG system does not replace this step. [1]

Prepare the concentrated Positive Control away from samples and keep completed qPCR reactions sealed. Include standards and controls when counting the 50-reaction kit capacity. [1]

Expected Outcome

A valid run produces a concentration-dependent standard series, reproducible technical wells and sample melt profiles consistent with the assay. Accepted sample quantities fall within the supported standard range and yield comparable titers after dilution correction.

Report physical genome titer in vg/mL together with the accepted curve range and relevant QC limitations. This measurement does not establish transduction efficiency, biological activity or the fraction of full-length genomes.

Troubleshooting

Observed problem	Check first	Suggested action
Sample is later than the lowest reliable standard	Standard range, added low-copy points and NTC background	Add lower Positive Control dilutions and test their reproducibility. Until supported, report below-range results or explicitly labeled extrapolations.
Low-copy standards amplify inconsistently	Serial-dilution mixing, transfers, replicates and NTC	Prepare a fresh dilution series and reassess low-end replicates. Do not force unreliable points into the curve to include a sample.
NTC or extraction blank has sample-like amplification	Reagents, standard handling and melt profile	Replace suspect aliquots and repeat setup in a clean area. A late Cq alone does not establish that low-copy sample signals are valid.
Standards fail to amplify	Mix, primer, template addition; program and read settings	Verify the 20 µL recipe, 60°C acquisition and reagent handling; remake the Positive Control series before interpreting negative samples.
Corrected titer changes with sample dilution	Dilution calculations, pipetting and possible inhibition	Repeat matched lysate dilutions at or above the recommended fivefold dilution. Select a reproducible range before reporting the original titer.
Unexpectedly high titer	DNase treatment and the fixed 200-fold extraction factor	Confirm incubation and inactivation records, check for contamination, and recalculate each factor once; repeat pretreatment if indicated.
Multiple melt peaks or large replicate differences	NTC, curve shape, bubbles, sealing and mixing	Inspect melt specificity and plate handling. Repeat affected wells or preparation; document exclusions instead of averaging discrepant signals.

Reference

[1] Beyotime. Universal AAV Titration Kit by qPCR, C2921S/C2921M. Version 2025.09.17; pp. 1-4: ITR scope, ROX compatibility, sample extraction, standards, cycling and titer calculation. Accessed 2026-10-03. Open source